SALVIN
Engineering - Processing - Packaging

MIX FRUIT-JAM PROCESSING PLANT

Your Trusted Partner for Consultancy and Turnkey Project Execution

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COMPANY OVERVIEW

Salvin Industries is a leading engineering, technology and consulting conglomerate, working closely with diversified industries spanning across food processing, agriculture, plastics, chemicals, metals healthcare, packaging, Dealer, distributer, super stocks, marketing plan, product analysis and other industries and business sectors offering complete turn-key solutions including advanced engineering, Industry 4.0 solutions, digital transformation and expert technical and management consultancy. With 10+ years of experience and global presence in 8+ countries, Salvin Industries is serving from the most demanding to the most distant customers across the globe. We empower them for rewriting their future with our depths of knowledge, unique ideas, technical expertise, creative solutions and enduring results, all at the most optimal costs, right at their doorsteps.

Armed with the eminence in technological expertise, global capabilities and a process driven approach, we have the ability to take on any challenging project. Crossing the geographical borders & cultures and living up to our core values Excellence, Respect, Care & Joy, we propel innovation to engineer solutions that build & foster equality. Working together as one team with all our business partners & associates, we are transforming industries, fueling economies, developing sustainable solutions and thereby elevating societies, globally. We realize the need to respond to the dynamism & speed with which the world is changing all around us and understand the need to anticipate

MIX FRUIT - JAM
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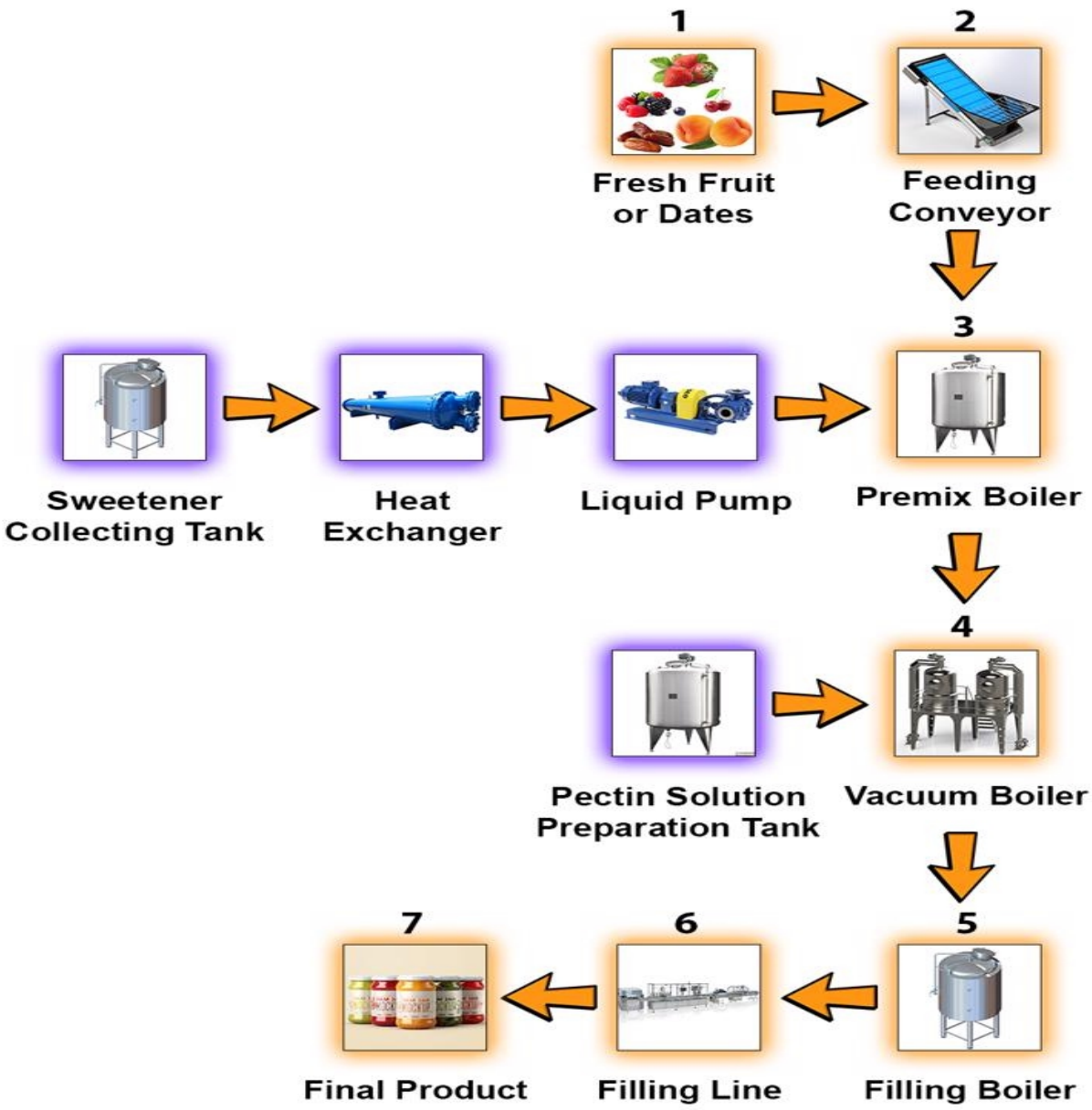
Strawberries, raspberries, blackberries, peaches, apricots, plums, or pineapples, you name it. All these fruits and many more, despite their differences, are suitable for jam processing.

Fruit jam processing can involve one single fruit or a combination of them. It all depends on the recipe. What's important if you want to achieve a perfect jam set is that there must be a balance of the three key elements: sugar, pectin and acids.

PROCESS FLOW MIX FRUIT-JAM MAKING PLANT



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THE PROCESS

Fruit Washing

Fresh fruit received from cold storage is tipped into a stainless steel receiving tank filled with recirculating water. Compressed air is introduced through a manifold at the bottom, creating thousands of rising bubbles that gently agitate the fruit and dislodge dirt, dust, and pesticide residues without bruising soft flesh. Overhead high-pressure spray nozzles provide a final rinse as fruit moves forward on the mesh conveyor. The water is continuously filtered and reheated to 30–40°C for optimal cleaning efficiency. This stage removes up to 95% of surface contaminants and prepares fruit for inspection.

Sorting & Inspection Conveyor

Cleaned fruit spreads into a single layer across a slow-moving PVC roller conveyor illuminated by bright LED lighting positioned at 45 degrees. This lighting angle reveals defects hidden in shadows. Trained visual inspectors stand on both sides and remove any fruit with mould spots, bird damage, insect injury, uneven ripening, or bruising. Alternatively, a colour sorting camera system with pneumatic ejectors can be installed for fully automated removal of defective pieces. Conveyor speed is adjustable between 3 to 8 metres per minute to match throughput. Only sound, premium fruit proceeds to the next stage.

Fruit Pitting & Crushing Unit

Whole fruit is metered into a machine fitted with rotating stainless steel paddles and stationary grids. For stone fruits like mango, peach, or apricot, a dedicated pitting section punches out the central seed with minimal flesh loss. For berry fruits, this stage simply crushes the skin. The fruit is reduced to 10–15mm chunks, releasing juice and breaking cell structure for efficient heat penetration during cooking. All contact parts are food-grade SS316 for acid resistance. The gap between rotor and screen is adjustable to control chunk size.

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THE PROCESS

Pulper / Refiner

Crushed fruit passes through a cylindrical sieve basket lined with fine stainless steel mesh. Rotating beater paddles or brushes spin at high speed, forcing soft pulp through the screen while skins, fibres, and hard fragments travel to a waste outlet. Interchangeable screens are available from 0.5mm for smooth jams to 2.0mm for rustic preserves. The machine is self-cleaning and can process up to 5 tonnes per hour. The extracted pulp is smooth, uniform, and ready for recipe formulation.

Preheater

Pulp is pumped through a jacketed tubular or scraped surface heat exchanger where it is rapidly raised to 85–100°C within 30 to 60 seconds. This sudden heating inactivates naturally occurring enzymes—particularly pectinase and polyphenol oxidase—that otherwise break down pectin and cause browning. Preheating also melts any crystallised sugars and dissolves pectin lumps if added at this stage. Precise temperature control (±1°C) prevents scorching. Hot pulp is held in an insulated buffer tank before formulation.

Formulation / Mixing Tank

Preheated pulp is discharged into steam-jacketed mixing kettles constructed from SS316, each vessel fitted with counter-rotating scraped-surface agitators. The paddles continuously sweep the heated walls, preventing caramelisation or burning. Granulated sugar, liquid glucose, pectin powder, citric acid, and buffering salts are added sequentially through a top manhole or vacuum suction li The recipe is stored in the PLC; ingredients are weighed and releas automatically. Batch temperature is maintained at 60–65°C during r to fully dissolve solids before vacuum cooking.

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THE PROCESS

Vacuum Evaporator / Cooking Kettle

The mixture is drawn into a vacuum evaporator operating at 600–650mmHg absolute pressure. Under vacuum, the boiling point of water drops to just 55–60°C. This gentle boiling evaporates moisture without subjecting the fruit to high temperatures, preserving volatile aroma compounds, natural fruit colour, and heat-sensitive vitamins. The Brix level is continuously monitored by an in-line refractometer; when the target of 65–68° Brix is reached, the vacuum is broken and the jam is discharged. Traditional open kettles are also available but vacuum evaporators produce superior quality.

Tube-in-Tube Sterilizer / UHT System

Hot jam is pumped through a multi-tube heat exchanger that raises the core temperature to 90–95°C and holds it for 30–60 seconds. This ultra-high temperature treatment destroys any remaining yeast, mould spores, and acid-tolerant bacteria that could cause fermentation or spoilage during shelf life. The narrow tubes ensure rapid, uniform heating without temperature gradients. If the sterilisation temperature falls below setpoint, a flow diversion valve automatically recirculates the product back to the holding tube. Sterilised jam proceeds directly to filling.

Jar Sterilizer / Autoclave

Empty glass jars arriving on pallets are transferred to a steam autoclave or continuous belt steriliser. Saturated steam at 121°C under positive pressure penetrates every cavity, killing all vegetative cells and bacterial spores. Jars remain in the sterilisation zone for 15–30 minutes depending on size and glass thickness. Lids and caps are sterilised separately in a rotating steam drum. Sterilised packaging is conveyed through a HEPA-filtered tunnel to the filling room, maintaining sterility until the moment of filling.

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Filling Machine

Sterilised hot jam at 85–90°C is delivered to a positive displacement piston filler or rotary auger filler designed specifically for high-viscosity, particulate-containing products. Each filling nozzle doses the exact volume with ±1g accuracy. Filling takes place in a laminar flow zone to prevent airborne contamination. The hot fill temperature also sterilises the inner surface of the jar. For oxygen-sensitive premium jams, a nitrogen gas flush displaces headspace air immediately before sealing.

Capping & Sealing Machine

Filled jars index into a capping turret. Caps are fed from a vibratory bowl sorter, oriented correctly, and placed onto jar mouths. A servo-driven chuck descends and applies precise, adjustable torque to achieve an airtight seal without damaging the cap or glass. The sealed jar then passes through an induction sealer or steam tunnel to create a vacuum seal as the jar cools. Vision inspection cameras check cap position, tamper band integrity, and fill level. Any defective jars are automatically rejected.

Cooling Tunnel / Labeling & Packaging

Jars enter a multi-stage cooling tunnel where fine water sprays gradually reduce the temperature from 85°C to 30–35°C over 15–25 minutes. Rapid cooling would cause thermal shock and glass breakage, so the temperature gradient is carefully controlled. After drying with high-velocity air knives, jars proceed to a rotary labelling station. Wrap-around or spot labels are applied with hot-melt adhesive. A laser jet or inkjet printer stamps the batch number, manufacturing date, and expiry date onto each label or jar. Finally, jars are automatically cartoned, palletised, and stretch-wrapped for dispatch.

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+91 97127 77034

www.salvinindustries.in

info.salvinindustries@gmail.com

210, Arved Transcube Mall,
Bandhu Nagar, Vijay Nagar,
Ranip, Ahmedabad, Gujarat 382480

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